SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct:

I P.HARINI (192521402) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Harini

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%	1	1	1	2
Debugging	25%	2	1	2	1
Optimization	20%	1	2	1	1
Analysis	20%	2	1	1	1
Code Quality	15%	1	1	1	1

25/30

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
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- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
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Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26. Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

Answer:

Given: IP = 192.168.10.65, Mask = 255.255.255.192 (/26), Gateway = 192.168.10.1. The workstation must communicate with the subnet 192.168.10.0/26.

Prefix /26 → host bits = 32 − 26 = 6 → block size = 2^6 = 64

Subnets of 192.168.10.x in steps of 64: .0, .64, .128, .192

Task 1: Subnet of the configured IP address

The last octet 65 lies between 64 and 127, so the address is in the subnet starting at 64.

Check by AND operation (last octet):

65 (IP) = 01000001

192 (mask) = 11000000

AND result = 01000000 = 64

The configured IP address 192.168.10.65 belongs to the subnet 192.168.10.64/26, not to 192.168.10.0/26.

Task 2: Network address, broadcast address and valid host range

Broadcast = Network + 64 − 1 Host range = Network + 1 to Broadcast − 1

Subnet	Network Address	Broadcast Address	Valid Host Range
192.168.10.0/26 (target subnet)	192.168.10.0	192.168.10.63	192.168.10.1 – 192.168.10.62
192.168.10.64/26 (subnet of workstation)	192.168.10.64	192.168.10.127	192.168.10.65 – 192.168.10.126

Task 3: Verification of the IP address and gateway

Item	Value	Belongs to subnet	Result
IP address	192.168.10.65	192.168.10.64/26	A valid host address of its own subnet, but not in the target subnet 192.168.10.0/26
Default gateway	192.168.10.1	192.168.10.0/26	Outside the subnet of the workstation, so it is not valid for this workstation

The gateway must be in the same subnet as the workstation (192.168.10.65 to 192.168.10.126 here). The gateway 192.168.10.1 is not in this range.

Task 4: Configuration errors

Error 1: The IP address is placed in the wrong subnet (192.168.10.64/26 instead of 192.168.10.0/26).

Error 2: The default gateway 192.168.10.1 is not in the same subnet as the workstation.

Because of Error 1, the workstation and the systems of 192.168.10.0/26 are in two different subnets, so they cannot communicate directly. Because of Error 2, the workstation cannot even reach its gateway, so no packet can be routed out of its own subnet. Hence communication fails.

Task 5: Corrected configuration

The workstation is moved into the target subnet 192.168.10.0/26 by selecting an unused host address from 192.168.10.1 – 192.168.10.62. The address .1 is kept for the router, so the workstation is given .10.

Parameter	Wrong Value	Corrected Value
IP address	192.168.10.65	192.168.10.10
Subnet mask	255.255.255.192	255.255.255.192
Default gateway	192.168.10.1	192.168.10.1

Check: 192.168.10.10 and 192.168.10.1 both lie in 192.168.10.0/26, neither is the network address (.0) nor the broadcast address (.63), and they are not the same. (If the workstation is meant to stay in 192.168.10.64/26, it would instead use the gateway 192.168.10.65 and a free address such as 192.168.10.66.)

Task 6: Usable host addresses after correction

Usable hosts = $2^6 - 2 = 64 - 2 = 62$

After correction the subnet 192.168.10.0/26 has 62 usable addresses. One address (192.168.10.1) is used by the router gateway, so 61 addresses remain for workstations.

2. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails. Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

Answer:

Given: Device A = 2001:db8::ff00:42:8329/64 and Device B = 2001:db8:0:0:1::1/64. An IPv6 address has 8 groups of 16 bits; in a /64 address the first 4 groups are the network prefix.

Task 1: Complete (expanded) notation

Rule: Add the zero groups hidden by :: and write every group with 4 hexadecimal digits.

Device A: 2001, db8, ff00, 42, 8329 = 5 groups written, so :: stands for 8 − 5 = 3 groups.

Device B: 2001, db8, 0, 0, 1, 1 = 6 groups written, so :: stands for 8 − 6 = 2 groups.

Device	Expanded Address
Device A	2001:0db8:0000:0000:0000:ff00:0042:8329/64
Device B	2001:0db8:0000:0000:0001:0000:0000:0001/64

Task 2: Network prefix of each device

Prefix /64 → take the first 64 bits = first 4 groups; the remaining 64 bits are the interface ID.

Device	First 4 groups (prefix)	Interface ID (last 4 groups)	Network Prefix
Device A	2001:0db8:0000:0000	0000:ff00:0042:8329	2001:db8::/64
Device B	2001:0db8:0000:0000	0001:0000:0000:0001	2001:db8::/64

Task 3: Same subnet verification

The first 64 bits of both addresses are identical (2001:0db8:0000:0000). Therefore Device A and Device B belong to the same subnet, 2001:db8::/64.

Task 4: Addressing or prefix mismatch

There is no mismatch in the /64 prefix itself, so the prefix length is not the cause of the failure. The problem is in the way Device B is written. In 2001:db8:0:0:1::1 the value 1 is the 5th group, which is bit position 65 to 80, so it is part of the interface ID and not of the subnet ID (4th group). It looks like an address of subnet 1 but it is actually still in subnet 0. The intended address is most likely 2001:db8:0:1::1/64, where the 1 is in the 4th group. Both devices also have irregular interface IDs, which makes the plan hard to check. If the prefix is correct, the remaining causes are outside the

addressing, such as the devices not being on the same link or VLAN, or ICMPv6 / neighbour discovery being blocked by an ACL or firewall.

Task 5: Optimized configuration

Both devices are placed in one clearly defined subnet, 2001:db8:0:1::/64 (subnet ID 1), with short and easily recognised interface IDs:

Device	Corrected Address	Network Prefix
Device A	2001:db8:0:1::10/64	2001:db8:0:1::/64
Device B	2001:db8:0:1::20/64	2001:db8:0:1::/64

Both addresses now have the same first 64 bits (2001:0db8:0000:0001), so they are in the same subnet and the plan is easy to verify.

Task 6: Available host addresses in the corrected subnet

Host bits = 128 − 64 = 64

Number of host addresses = 2^{64} = 18,446,744,073,709,551,616

IPv6 has no broadcast address, so the "− 2" of IPv4 is not used.

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

Answer:

Given: Network = 192.168.1.0/24 (256 addresses), divided using /26, /27 and /28, one subnet for each department.

Assumption: The question does not give the number of hosts needed by each department, so the following requirements are assumed to explain the problem. The method is the same for any other values.

Department	Mask given	Hosts required (assumed)	Subnets allocated in order
Department A (Administration)	/26	25	1st
Department B (Accounts)	/27	28	2nd
Department C (Library)	/28	20	3rd

Block size: /26 = 2^6 = 64, /27 = 2^5 = 32, /28 = 2^4 = 16

The subnets are placed one after the other starting from 192.168.1.0.

Task 1: Network address, broadcast address and host range

Dept A: 192.168.1.0 + 64 − 1 = 192.168.1.63

Dept B: starts at 192.168.1.64; 64 + 32 − 1 = 95, so 192.168.1.95

Dept C: starts at 192.168.1.96; 96 + 16 − 1 = 111, so 192.168.1.111

Department	Subnet	Network	Broadcast	Host Range
Dept A	192.168.1.0/26	192.168.1.0	192.168.1.63	192.168.1.1 – 192.168.1.62
Dept B	192.168.1.64/27	192.168.1.64	192.168.1.95	192.168.1.65 – 192.168.1.94
Dept C	192.168.1.96/28	192.168.1.96	192.168.1.111	192.168.1.97 – 192.168.1.110

Task 2: Usable and unused IP addresses

Usable = $2^h - 2$ Unused = Usable − Hosts required

Dept A: $2^6 - 2 = 62$ usable; 62 − 25 = 37 unused

Dept B: $2^5 - 2 = 30$ usable; 30 − 28 = 2 unused

Dept C: $2^4 - 2 = 14$ usable; 14 − 20 = −6, so the subnet is short by 6 addresses

Addresses allocated = 64 + 32 + 16 = 112; Still free in the /24 block = 256 − 112 = 144 (192.168.1.112 to 192.168.1.255)

Task 3: Inefficient allocations

Dept A: /26 gives 62 hosts for a need of 25, so 37 addresses (about 60%) are wasted.

Dept C: /28 gives only 14 hosts for a need of 20, so the department has insufficient addresses.

Dept B: /27 gives 30 hosts for a need of 28, which is efficient.

Task 4: Allocation mistakes

Mistake 1: The masks were chosen in a fixed order (/26, /27, /28) and not according to the number of hosts in each department.

Mistake 2: The largest block was given to Dept A, which needs fewer hosts than Dept B and Dept C.

Mistake 3: Free space (144 addresses) was left unused while Dept C ran out of addresses.

Task 5: Optimized scheme using VLSM

In VLSM the subnets are arranged from the largest host requirement to the smallest, and each one gets the smallest block that fits. The needs are 28, 25 and 20 hosts. The smallest h with $2^h - 2 \geq 28$ (and also ≥ 25 and ≥ 20) is h = 5, so each department needs a /27 (30 usable hosts).

Department	Subnet	Network	Broadcast	Host Range
Dept B (28 hosts)	192.168.1.0/27	192.168.1.0	192.168.1.31	192.168.1.1 – 192.168.1.30
Dept A (25 hosts)	192.168.1.32/27	192.168.1.32	192.168.1.63	192.168.1.33 – 192.168.1.62
Dept C (20 hosts)	192.168.1.64/27	192.168.1.64	192.168.1.95	192.168.1.65 – 192.168.1.94

Spare addresses: Dept B = 30 − 28 = 2, Dept A = 30 − 25 = 5, Dept C = 30 − 20 = 10

Addresses used = 3 × 32 = 96; free block kept for growth = 256 − 96 = 160 (192.168.1.96 to 192.168.1.255)

Result: No department is short of addresses now, the wasted addresses inside the subnets are reduced from 39 to 17, and a free block of 160 addresses remains for new departments.

4. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails. Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

Answer:

Given: Device A = 2001:db8::ff00:42:8329/64 and Device B = 2001:db8:0:0:1::1/64.

Task 1: Complete (expanded) notation

Device A: :: replaces 8 − 5 = 3 zero groups. Device B: :: replaces 8 − 6 = 2 zero groups.

Device	G1	G2	G3	G4	G5	G6	G7	G8
A	2001	0db8	0000	0000	0000	ff00	0042	8329
B	2001	0db8	0000	0000	0001	0000	0000	0001

Device A = 2001:0db8:0000:0000:0000:ff00:0042:8329/64

Device B = 2001:0db8:0000:0000:0001:0000:0000:0001/64

Task 2: Network prefix of each device

For /64, the prefix = G1 to G4 (64 bits).

Device A: 2001:0db8:0000:0000 → 2001:db8::/64

Device B: 2001:0db8:0000:0000 → 2001:db8::/64

Task 3: Same subnet verification

G1 to G4 are identical in both addresses, so both devices are in the same subnet 2001:db8::/64.

Task 4: Addressing or prefix mismatch

The /64 prefixes match, so the prefix length is not the fault. In Device B, the value 1 is in G5, which belongs to the interface ID (bits 65 to 80), and not in G4, the subnet ID. So 2001:db8:0:0:1::1 is not an address of subnet 1 as it appears; it is a host address of subnet 0 with an unusual interface ID. The probable error is that the address was written with an extra group, instead of 2001:db8:0:1::1. Since the addressing is consistent, other causes such as the link or VLAN, an ACL / firewall blocking ICMPv6, or neighbour discovery problems should also be checked using ping and the neighbour table in Packet Tracer.

Task 5: Optimized configuration

Device	Corrected Address	Network Prefix
Device A	2001:db8:0:1::10/64	2001:db8:0:1::/64
Device B	2001:db8:0:1::20/64	2001:db8:0:1::/64

Both addresses have the same first four groups (2001:0db8:0000:0001) and short interface IDs, so the plan is clear and easy to verify.

Task 6: Available host addresses in the corrected subnet

Host bits = 128 − 64 = 64

Host addresses = 2^{64} = 18,446,744,073,709,551,616

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended network. Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

Answer:

Given: Destination IP = 192.168.2.100. The default route is the route 0.0.0.0/0.

Task 1: Longest-prefix match

The router compares the destination with every entry and selects the matching entry that has the longest prefix:

Routing Entry	Address Range	Match?	Prefix length
192.168.1.0/24	192.168.1.0 – 192.168.1.255	No	–
192.168.2.0/24	192.168.2.0 – 192.168.2.255	Yes	24
Default (0.0.0.0/0)	All addresses	Yes	0

Two entries match, but /24 is longer than /0, so the entry 192.168.2.0/24 is selected.

Task 2: Next-hop router

The selected entry 192.168.2.0/24 gives the next hop 10.0.0.3. The packet for 192.168.2.100 is therefore forwarded to the router 10.0.0.3, and not to 10.0.0.1 (default).

Task 3: Destination belongs to the intended subnet?

192.168.2.100 AND 255.255.255.0 = 192.168.2.0

Yes. 192.168.2.100 is inside 192.168.2.0/24 (network 192.168.2.0, broadcast 192.168.2.255, hosts .1 to .254), and it is a valid host address.

Task 4: Routing configuration errors

The matching and the subnet check are both correct, so the table forwards the packet exactly as written, and the fault must be in the details of the table. The following points need correction:

1. The next hop 10.0.0.3 may be wrong or unreachable (for example, the next hops of the two networks may have been swapped, or 10.0.0.3 has no route back or forward to 192.168.2.0/24).
2. The table does not show the exit interface or the mask of the 10.0.0.x link, so it cannot be verified that all three next hops are on a directly connected network.
3. The default route is written as the words “Default Route”; it must be configured as 0.0.0.0/0.

These can be checked in Packet Tracer using show ip route, ping 10.0.0.3 and traceroute 192.168.2.100.

Task 5: Optimized routing table

The two /24 routes have different next hops, so they cannot be combined into one summary route; the specific routes are kept and the default route is written correctly. Each route is checked so that its next hop is reachable on a connected interface.

Destination	Mask	Next Hop	Type
192.168.1.0	255.255.255.0	10.0.0.2	Static (specific)
192.168.2.0	255.255.255.0	10.0.0.3	Static (specific)
0.0.0.0	0.0.0.0	10.0.0.1	Default route

```
Router(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.2
Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.3
Router(config)# ip route 0.0.0.0 0.0.0.0 10.0.0.1
```

Task 6: Route used when no destination matches

If the destination does not match 192.168.1.0/24 or 192.168.2.0/24, the default route 0.0.0.0/0 is used and the packet is sent to the next hop 10.0.0.1.

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